Does match workload affect tennis win probability?

More time on court was associated with lower win odds. Adding workload did not improve held-out predictions.

Data through 22 October 2026. Forecasts dated 23 October 2026. English edition.

Fatigue is approximated by prior match workload, not measured directly. The findings show association, not a causal effect.

Data coverage

The source snapshot contained **5,055 records**, reduced to **5,054 unique 2026 records**. Of these, **4,874** had standard completed singles scores. This is a partial-season source snapshot, not a complete 2026 season census.

Tour	Completed records	With duration	Median duration
ATP	2,548	2,467	111 minutes
WTA	2,326	2,042	99 minutes

Counts include singles played in team events. Retirements, walkovers and non-standard scores are excluded. Duration medians use records with reported duration.

What the models use

Baseline: pre-match ranking, age, qualifier/lucky-loser status, and rank interactions with tour, surface, best-of-five format and late rounds. **Workload model:** adds the difference between players in prior hours and prior match counts within the same event.

Regularized logistic regression was tuned on a validation set. Only earlier rounds contribute workload. The target match score and duration are excluded from predictors. Player ordering is independent of the outcome; mirrored training makes probabilities complementary when players are swapped.

Evaluation design

Eligible matches	Training	Validation	Test	Boundary exclusions
1,578	887	187	174	330

Training uses January-June; validation July-August; test events begin in September or later. A conservative 21-day event boundary buffer addresses mixed match-day and tournament-start dates. The test set covers 16 events. After evaluation, the forecast models were refitted on all 1,578 eligible matches.

Association without proven prediction improvement

Adjusted workload association

An extra hour of prior same-event play relative to the opponent was associated with an odds ratio of **0.894** (95% event-bootstrap interval: **0.825-0.984**), after adjustment for the model features.

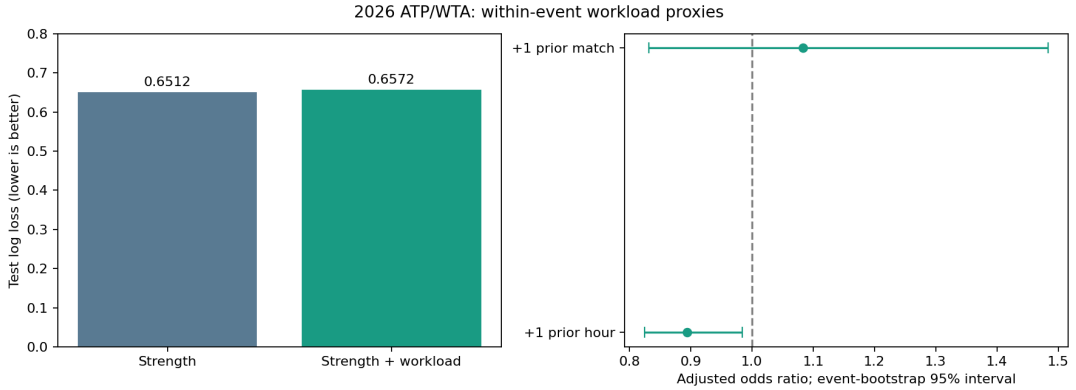
This is about **10.6% lower odds**, not a 10.6 percentage point decrease in win probability. The prior-match-count odds ratio was 1.084 (95% interval: 0.832-1.483). These association estimates use the development set; they do not establish causation.

Independent test results: 174 matches

Model	Accuracy	AUC	Log loss	Brier score
Baseline	60.9%	0.671	0.6512	0.2297
Baseline + workload	60.9%	0.660	0.6572	0.2323

Higher AUC is better. Lower log loss and Brier score are better. Both models achieved 60.9% accuracy.

Adding workload changed log loss by +0.0060 (95% interval: -0.0095 to +0.0272). There is no robust evidence of improved predictive performance.



Intervals resample whole events: 500 bootstrap repetitions for adjusted coefficients and 3,000 for the test log-loss comparison. The independent test sample is small.

How to interpret the result

Time on court is an incomplete fatigue proxy. Previous opponent strength, injury, travel, sleep, training and scheduling can affect both workload and outcomes. Qualifying and doubles workload are not fully captured. Excluding retirements and missing durations can introduce selection bias.

Precise rest-day features were not calculated because source dates have inconsistent meanings. The evidence supports a workload association in this sample, but does not justify preferring the workload model for future predictions.

Frozen forecasts for 23 October 2026

Fixtures and current rankings were checked against TennisStats. Prior same-event minutes come from TennisMyLife. Each probability below refers to **Player A**.

Event / round	Player A / Player B	Prior minutes A / B	A wins Baseline	A wins + workload
Lyon ATP QF	Corentin Moutet / Tommy Paul	289 / 126	27.4%	28.5%
Lyon ATP QF	Flavio Cobolli / Brandon Nakashima	68 / 210	64.3%	64.5%
Brussels ATP QF	Tallon Griekspoor / Jiri Lehecka	243 / 79	32.8%	32.3%
Ningbo WTA QF	Iva Jovic / Diana Shnaider	82 / 261	53.1%	57.8%

QF = quarterfinal. These are dated research forecasts with history frozen at 22 October, conditional on a normal completed match. Differences between columns come from separately refitted models and are not causal fatigue effects. Refresh the data before using them for later matches.

TennisStats integration

The official demo match (event 971951) was uniquely matched to TennisMyLife, with the winner, two sets and 22 games verified. Four public fixtures were matched for prediction. The client supports account export, pagination and secret-based credentials; full TennisStats historical export was not run because it requires an account API key.

Kaggle execution

Open the executed Kaggle notebook

Private notebook, version 1 | CPU | Successful run: **55.7 seconds** | Run ID: 361288222

The downloaded cloud notebook contains six executed code cells with no errors. Cloud metrics match the local results to an absolute tolerance of 10^{-12}. The frozen snapshot can run without internet. Optional account export reads TENNISSTATS_API_KEY from Kaggle Secrets.

Validation covered score parsing, prevention of current/future-round workload leakage, probability symmetry when players are swapped, and the official API demo schema.

Sources and fixture references

TennisMyLife data and field documentation
TennisStats public schedules | Official API documentation

Fixture checks: Moutet / Paul | Cobolli / Nakashima | Griekspoor / Lehecka | Jovic / Shnaider